

VERISILICON Macro GPIO Generator Spec

Rev:1.1

VeriSilicon Microelectronics (Shanghai) Co., Ltd.

Checked by: _____

Approved by: _____

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VeriSilicon Microelectronics (Shanghai) Co., Ltd.

20F, No.560, Songtao Road, Zhangjiang Center, Zhangjiang Hi-Tech Park, Pudong New Area, Shanghai 201203, PRC

Tel : +86-21-5131-1118

Fax : +86-21-5133-4119

Web : <http://www.verisilicon.com>

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1. General Description

The GPIO consists of 32 external data interfaces from/to I/O pads and auxiliary hardware data interfaces to/from auxiliary data target/sources. An interface is provided to configure arbitrary number of pins for interrupt sources and interrupt controller for each data stream. The module is scalable to multiple instantiations to facilitate wider bandwidth transfer for I/O.

2. Features List

- Compliance with the AMBA2.0 APB interface;
- Configurable input/output pins number up to 32, setup by software or hardware;
- Alternative control modes set by software or hardware;
- After reset, all the GPIO lines default direction and the output data are configured ;
- Configurable reset values data output to PAD on software mode;
- Programmable interrupt generation capability on any number of pins;
- Edge sensitive conditional interrupts configurable on each GPIO in pins.
- Debounce circuits can be embedded for data line from 0 to N;
- Each pin can be uniquely setup in different modes;
- The definition of filtered glitch is programmable;

3. Application System Circuit

Figure 3-1 shows the GPIO application circuit in the system(N=31)

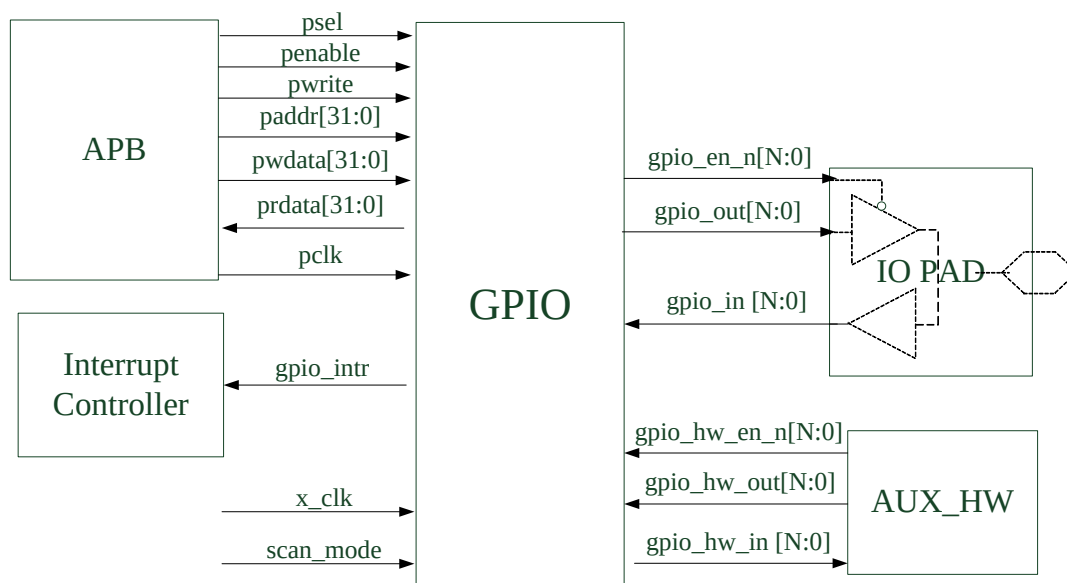


Figure 3-1 GPIO Application Circuit

4. Block Diagram

4.1. Block diagram

Figure 4-1 shows the GPIO block structure diagram.

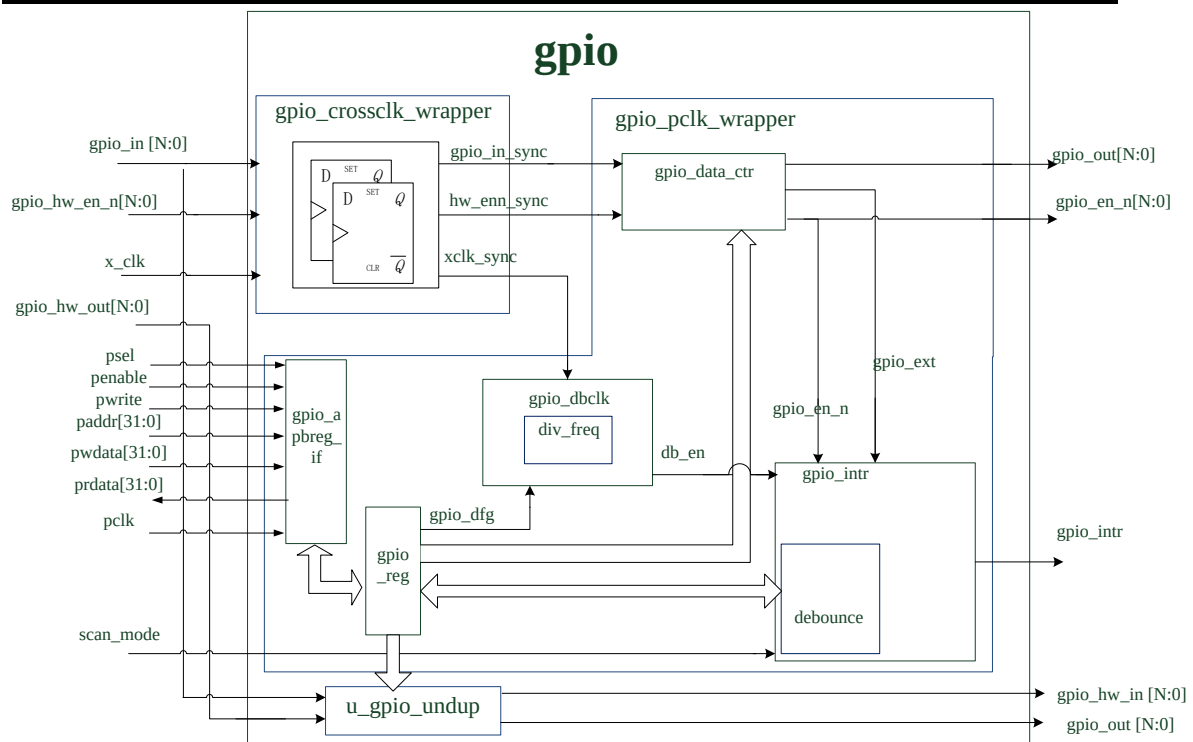


Figure 4-1 GPIO Block Diagram

4.2. Block DESCRIPTION

4.2.1. Overview

The gpio controls the output data and direction of external I/O pads. It also can read back the data on external pads using memory-mapped registers. The APB interface accesses the registers to read/write data and status, the data_ctrl block decides the data flow according to software or hardware setup, and intr block compare data to defined values. To avoid mistake, the dbclk block is used to eliminate the bounce generated in the setup phase.

4.2.2. gpio_crossclk_wrapper

This block is used as a synchronization signal.

4.2.3. gpio_apbreg_if

This block converts APB interface to similar with C51.

4.2.4. gpio_reg

This block provides initialization process and access of data and status for software.

4.2.5. gpio_undup

This block controls the data flow as set by software or hardware.

4.2.6. gpio_data_ctrl

This block controls the data direction as set by software or hardware.

4.2.7. gpio_intr

This block captures the data line, and generates interrupt status.

4.2.8. gpio_dbclk

This block defines the debounce clock as setup.

5. Pin List and parameter list

Table 5-1 is the GPIO pin list.

Table 5-1 GPIO Pin List

Name	I/O	Description
p_sel	I	Input from APB selection signal
p_enable	I	Input from APB enable signal
p_write	I	Input from APB write signal
p_addr [31:0]	I	Input from APB address. only p_addr[5:2] is used, the other bits are unused.
p_wdata [31:0]	I	The APB data which written in to GPIO
p_rdata [31:0]	O	The APB data which read out from GPIO
p_resetn	I	Low active, APB reset signal.is asynchronously asserted and synchronously de-asserted to p_clk. synchronization must be external to this component.
p_clk	I	The APB clock to synchronize GPIO.
gpio_hw_en_n[31:0]	I	Low active, enable signal input from hardware.
gpio_hw_out[31:0]	I	Hardware data input
gpio_hw_in [31:0]	O	Hardware data output
gpio_en_n[31:0]	O	Low active, enable gpio output data to Pad.
gpio_out[31:0]	O	The data output to PAD.
gpio_in[31:0]	I	The data input from PAD, To ensure accurate data transmission in software mode, it is essential that the data validity period exceeds five PCLK cycles.
gpio_intr	O	Interrupt flag for special GPIO lines
x_clk	I	The external clock for debounce. less than $1/3f_{pclk}$
scan_mode	I	Used for scan chain.

Table 5-2 is the GPIO parameter list.

Table 5-2 GPIO Parameter List

PARAMETER NAME	Default Value	DESCRIPTION	EFFECTIVE VALUE
GPIO_OUT_RESET_VALUE	0x0	This parameter configures the default value of GPIO_OUT after power-on reset. the bit width is the same as the GPIO_BUS_WIDTH during configuration.	{multi}{GPIO_BUS_WIDTH}{0x0, 0x1}
GPIO_DIR_DEFAULT	0x0	This parameter configures the default direction of GPIO_DIR register after power-on reset. the bit width is the same as the	{multi}{ GPIO_BUS_WIDTH }{0x0, 0x1}

		GPIO_BUS_WIDTH during configuration.	
GPIO_BUS_WIDTH	0xFFFF FFFF	This parameter configures the default number of gpio pins.	0x1,.....0xFFFF_FFFF
GPIO_DEBOUNCE_WIDTH	0xFFFF FFFF	This parameter configures the default number of debounce pins, must less than or equal to GPIO_BUS_WIDTH .	0x0,0xFFFF_FFFF

6. REGISTER FILE AND MEMORY MAPPING DESCRIPTION

6.1.REGISTER AND MEMORY MAPPING SUMMARY TABLE

Table 6-1 GPIO Register Memory Map

Name	Address	Width	Description
gpio_dir	00	N	Direction of data flow register
gpio_ctrl	04	N	Control mode select register
gpio_set	08	N	output data set register
gpio_clr	0C	N	output data clear register
gpio_odata	10	N	output data register
gpio_idata	14	N	input data register
gpio_ien	18	N	Interrupt enable register
gpio_is	1C	N	Interrupt sense register
gpio_ibe	20	N	Interrupt both edge register
gpio_iev	24	N	Interrupt event register
gpio_ris	28	N	Raw interrupt status register
gpio_im	2C	N	Interrupt mask register
gpio_mis	30	N	Masked interrupt status register
gpio_ic	34	N	Interrupt clear register
gpio_db	38	N	Delete Bounce flag
gpio_dfg	3C	32	Define debounce clock
N is configurable ,which is equals to gpio_bus_width(1~32).			

6.2.REGISTER DESCRIPTION

6.2.1. gpio_dir(Offset: 0x00)

Bits	Access	Default	Description
[N-1:0]	RW	GPIO_DIR_DEFAULT	Each bit mapping to the gpio_en_n and control the direction of data flow in software mode: "0" for input from gpio_in, "1" for output to gpio_out. the value of reset release can be configured.

6.2.2. gpio_ctrl(Offset: 0x04)

Bits	Access	Default	Description
------	--------	---------	-------------

[N-1:0]	RW	N'h0	"0" for software mode, each bit of gpio_dir is used to setup the direction for corresponding pin; "1" for hardware mode, auxiliary hardware signal, gpio_hw_en_n is used to configure the direction for the corresponding pin.
---------	----	------	--

6.2.3. gpio_set(Offset: 0x08)

Bits	Access	Default	Description
[N-1:0]	WO	N'h0	GPIO output bit set register. If writing high to each bit, the corresponding output pins would change to HIGH and other pins maintain the original output when the pin direction is configured as output. When writing low or the pin direction is input, writing the register has no effective.

6.2.4. gpio_clr(Offset: 0x0C)

Bits	Access	Default	Description
[N-1:0]	WO	N'h0	GPIO output bit clear register. If writing high to each bit, the corresponding output pins would change to LOW and other pins maintain the original output when the pin direction is configured as output. When writing low or the pin direction is input, writing the register has no effective.

6.2.5. gpio_odata(Offset: 0x10)

Bits	Access	Default	Description
[N-1:0]	RW	GPIO_OUT_RESET_VALUE	GPIO output pins mapping register and each bit is valid when the corresponding pin direction is configured as output in software mode. Writing access will set the external pins directly and reading access will get the external pins value.

6.2.6. gpio_idata(Offset: 0x14)

Bits	Access	Default	Description
[N-1:0]	RO	N'h0	GPIO input pins mapping register and each bit is valid when the corresponding pin direction is configured as input.

6.2.7. gpio_ien(Offset: 0x18)

Bits	Access	Default	Description
[N-1:0]	RW	N'h0	For each bit, "0" disable the interrupt event on the corresponding pin, while "1" to enable the corresponding pins trigger interrupt.

6.2.8. gpio_is(Offset: 0x1C)

Bits	Access	Default	Description
[N-1:0]	RW	N'h0	For each bit, "0" is edge-detecting and "1" is level-sensitive on corresponding pin.

6.2.9. gpio_ibe(Offset: 0x20)

Bits	Access	Default	Description
[N-1:0]	RW	N'h0	For each bit, "0" is single edge-sensitive (either falling edge or rising edge), decided by the bit of gpio_iev. "1" is both edges on corresponding pin trigger an interrupt.

6.2.10. gpio_iev(Offset: 0x24)

Bits	Access	Default	Description
[N-1:0]	RW	N'h0	For each bit, "0" means that interrupt is triggered in falling edge/low level of corresponding pins, "1" means interrupt is triggered in rising edge/high level of corresponding pins. If the gpio_is and gpio_iev is asserted at the same time, the detection is level interrupted. If the gpio_db and gpio_iev is asserted at the same time, the high level burrs are filtered.

6.2.11. gpio_ris(Offset: 0x28)

Bits	Access	Default	Description
[N-1:0]	RO	N'h0	Interrupt Pending status register. only when IO pad as input mode will happen. "1": Interrupt pending "0": Interrupt not pending

6.2.12. gpio_im(Offset: 0x2C)

Bits	Access	Default	Description
[N-1:0]	RW	N'h0	For each bit, "0" means that the interrupt on the corresponding gpio_in pins is unmasked while "1" is masked.

6.2.13. gpio_mis(Offset: 0x30)

Bits	Access	Default	Description
[N-1:0]	RO	N'h0	For each bit, this is the masked interrupt status on the corresponding gpio_ris bit. For example, if gpio_in pins generate interrupt, gpio_ris register could read the interrupt information, but if masked this interrupt by gpio_im register, the gpio_mis register will not know the interrupt information.

6.2.14. gpio_ic(Offset: 0x34)

Bits	Access	Default	Description
[N-1:0]	WO	N'h0	For each bit, write "1" to clear the edge-sensitive interrupt status on the corresponding pin, write "0" to a bit has no effects on the corresponding interrupt.

6.2.15. gpio_db(Offset: 0x38)

Bits	Access	Default	Description
------	--------	---------	-------------

[N-1:0]	RW	N'h0	For each bit, write "1" to enable the function of debounce for each line, write "0" to disable the debounce for each line. This parameter must be consistent with GPIO_DEBOUNCE_WIDTH so that the debounce function can be used normally.
---------	----	------	---

6.2.16. gpio_dfg(Offset: 0x3C)

Bits	Access	Default	Description
[31:0]	RW	32'h0	If the value is 32'h0000_0000, the db_clk, which is used to define the glitch, will be pclk. If the value is 32'hFFFF_FFFF, the db_clk will be x_clk. Otherwise the db_clk frequency will be pclk/(gpio_dfg+1). Anyway, it has no effect on debounce when gpio_db is 0. NOTE: the width of this register is fixed.

7. Function Description

7.1. Data and Control Flow

The GPIO controls the output data and direction of external I/O pads. It also can read back the data on external pads using memory-mapped registers.

There are two methods for generating the default source of the input data, the output data, and the control of each signal—through software or hardware control. Figure 7-1 shows the GPIO block data flow. Software control occurs over the APB bus interface; hardware control is through the auxiliary hardware control interface. Figure 7-2 shows the GPIO data control flow.

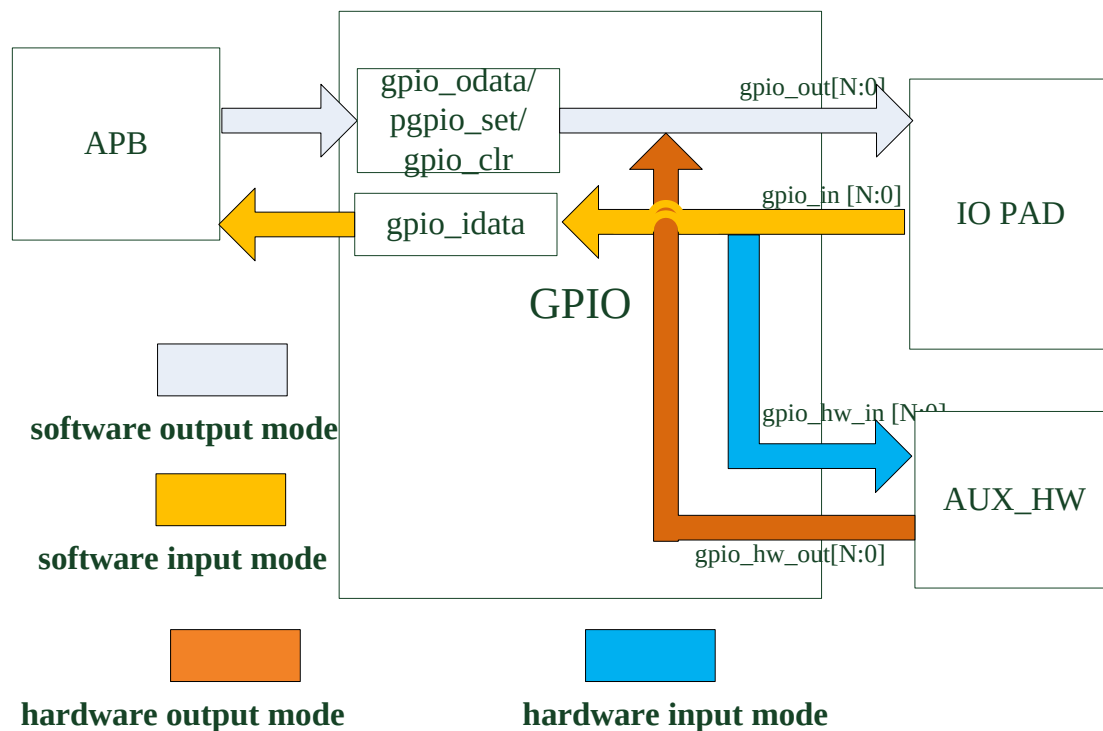


Figure 7-1 GPIO block data flow

32'h0000_0000	pclk, as system clk	
32'hFFFF_FFFF	x_clk, as external clk	
Any other value else	pclk/(gpio_dfg+1)	

There is only ONE db_clk for each GPIO implementation.

If the debounce function (gpio_db) is enabled but the corresponding debounce channel (GPIO_DEBOUNCE_WIDTH) is not opened, it indicates an incorrect configuration. In this case, the data sent for interruption without debounce will be four clock cycles after the GPIO input has been synchronized.

8. Interface Timing Diagram

This section illustrates the interface timing diagrams.

8.1. APB INTERFACE Write Transfer

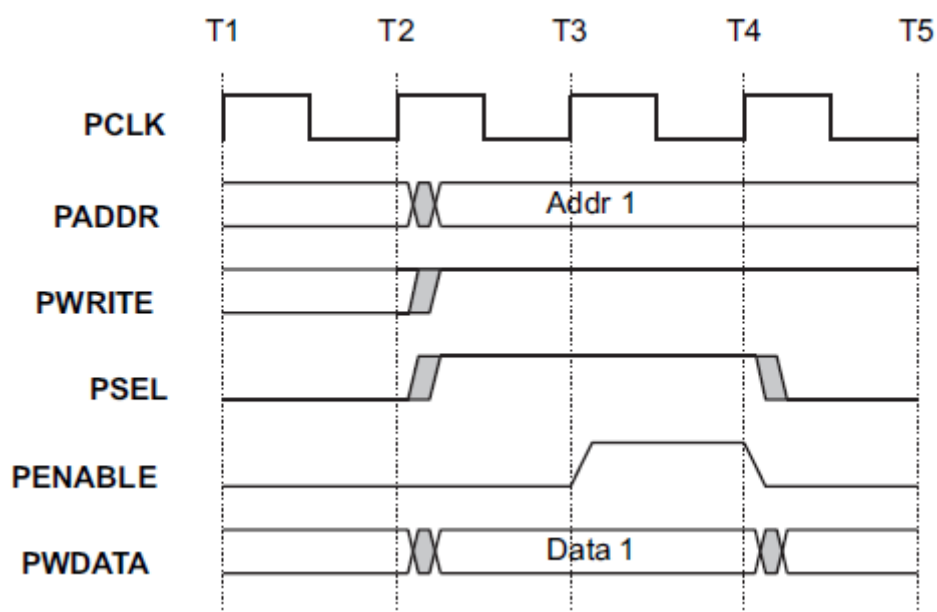


Figure 8-1 APB write transfer

8.2. APB INTERFACE Read Transfer

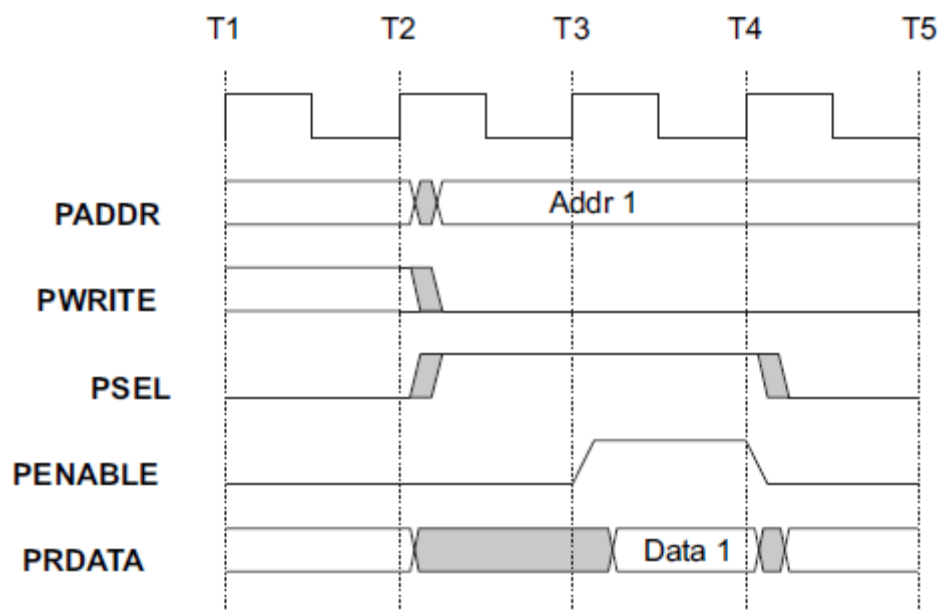
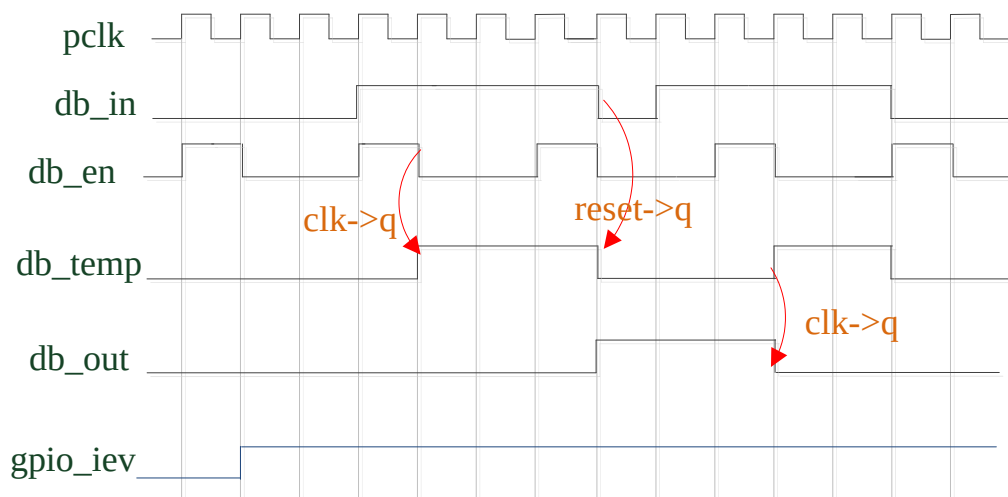


Figure 8-2 APB read transfer

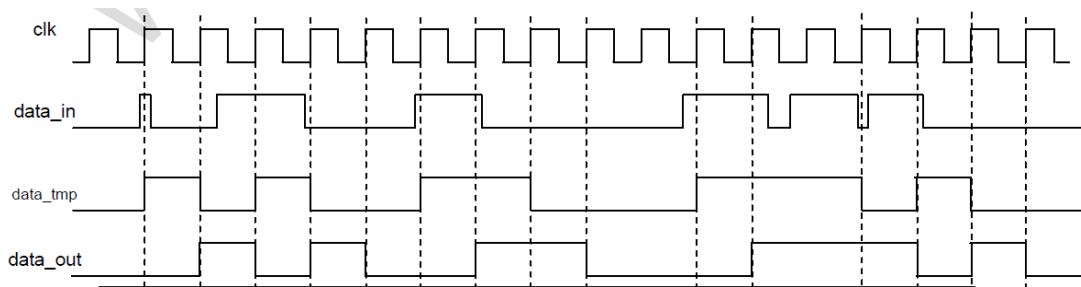
For the RO(Read Only) register, the writing operation has no effect on the value of the register. As for the WO(Write Only) register, the writing operation only affects register and/or system state.

8.3. DEBOUNCE TIMING DIAGRAM



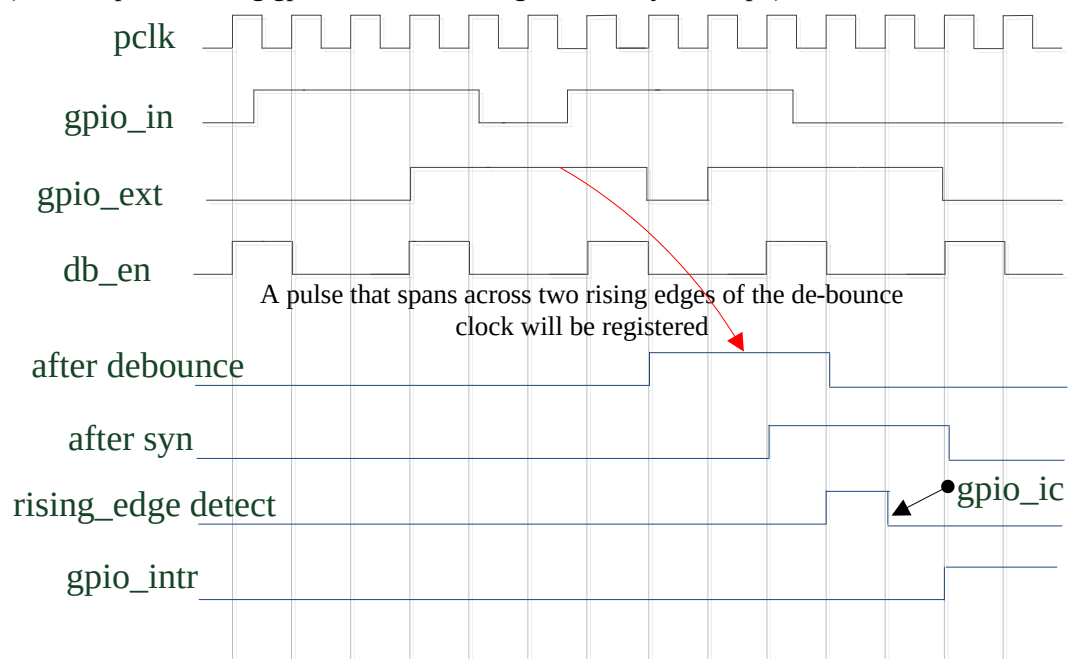
It is obvious that this block will filter the bounce. While the signal latched twice or more, keeping valid more than one clock pluses, will be passed and delayed for sync output.

8.4. META-STABILITY SYNC TIMING DIAGRAM



8.5. INTERRUPT TIMING DIAGRAM

(for example in writing `gpio_ic` to clear all edge-sensitivity interrupt.)



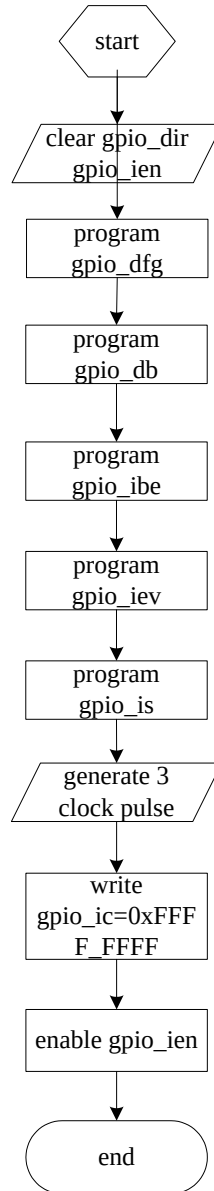
Pulses that are not picked up to 2 times by the rising edge of the `db_en` are filtered out, leaving only the last pulse to generate an interrupt.

9. Configuration Program Flow

This section elaborates on how to configure program flows.

9.1.SETUP THE INTERRUPT

To work in edge-sensitive interrupts mode, the software must perform the following initialization sequence to prevent spurious interrupt by the system:



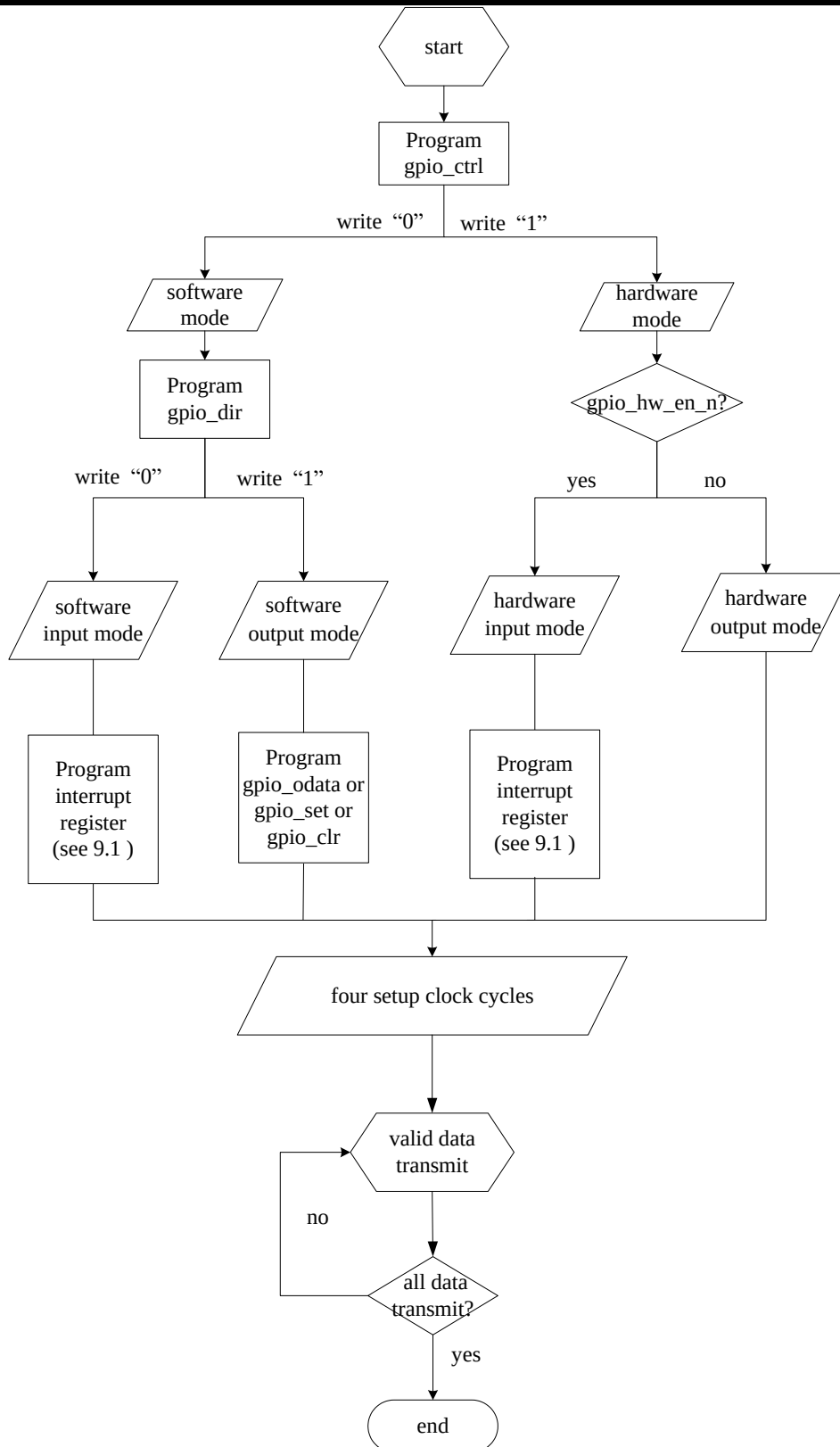
1. To trigger interrupt in setup condition, it is necessary to clear gpio_dir in all modes;
2. It is strongly recommended to clear gpio_ien to avoid the abnormality;
3. Program the "gpio_dfg" and define Filtered Glitch to select the source for db_clk;
4. Program "gpio_db" to enable or disable debounce the glitch;
5. Program the register: "gpio_ibe", appropriately as individual or both-edge detection;
6. Program the register: "gpio_iev", if have selected individual edge transactions previously;
7. Program the register: "gpio_is", to select edge-triggered path;
8. Generate three clock pulses to clean interrupt pipeline;
9. Program the register "gpio_ic" value = 0xFFFF_FFFF to clear all edge-detected interrupts;
10. Program the register "gpio_ien" to enable the interrupts at last. And ensure that the

"gpio_ien" register is configured at least five cycles after the configuration of the "gpio_iev" register to avoid generate unnecessary interrupts.

11. Make sure that the 'gpio_in' pin remains stable during the above operation.

9.2.SETUP THE DATA FLOW

1. Program the register:
2. Write "gpio_ctrl" to select the control mode: write "0" software control; 1: hardware control.
3. In software mode, program the register "gpio_dir" to control the direction of data stream for each pin. Writing "0" to gpio_dir indicates that the data flow is from gpio_in to apb, signifying software input mode. Conversely, writing "1" signifies that the data flow is from apb to pad, indicating software output mode. In hardware mode, if gpio_hw_en_n is asserted, the data flow is from gpio_in to gpio_gpio_hw_in, representing hardware input mode. On the other hand, if gpio_hw_en_n is deasserted, the data flow is from gpio_hw_out to gpio_out, indicating hardware output mode.
4. Maintain the state of gpio_in, gpio_hw_out, and gpio_hw_en_n unchanged for four setup clock cycles after the configuration above when beginning to transmit valid data. (2 debounce clock cycles and 2 meta-stability clock cycles).



10. APPLICATION NOTE

10.1. CLOCK AND RESET SCHEME

The asynchronous reset signal is used for the rst_n of DFF.

The debounce block uses asynchronous signal to delete the bounce like db_clk. The db_clk and system clock can be asynchronous. But db_clk and meta-stability clock must be synchronous. The clock and reset of GPIO use the signal of APB. The clock frequency of debounce can be either clock frequency of APB or other clock frequencies. for example, x_clk. The following block diagram shows the synchronization of the x_clk signal.

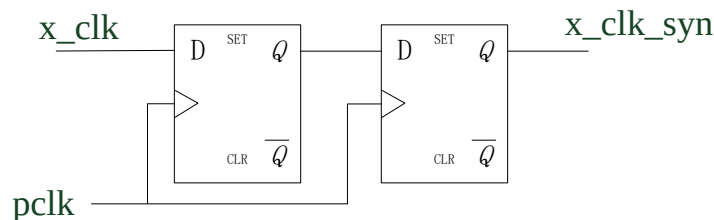
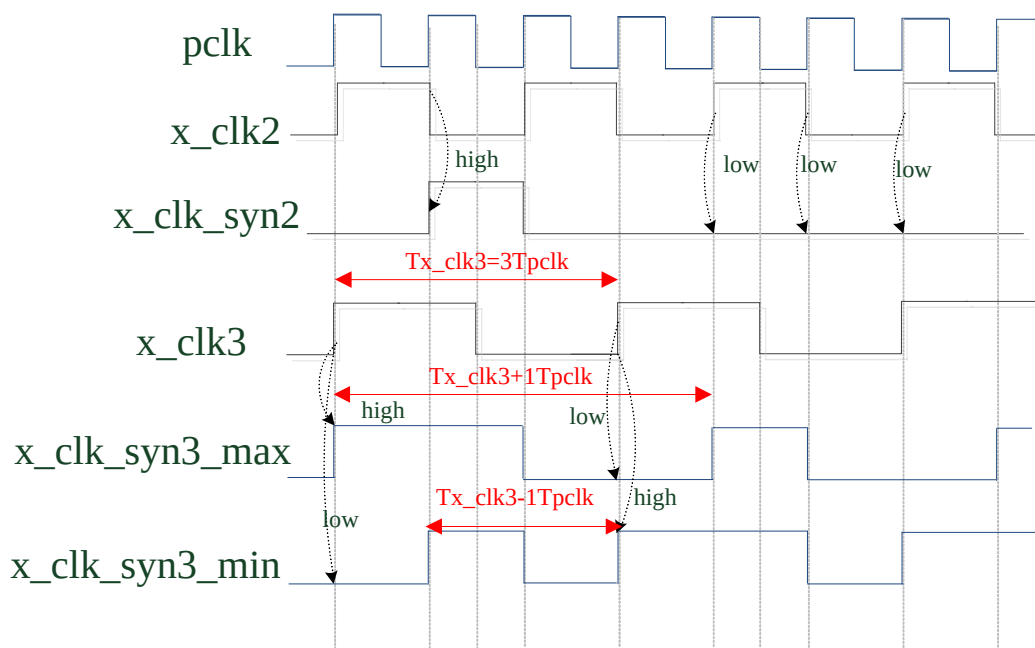


Figure 10-1 Synchronization of the xclk Signal

The selected frequency of x_clk should be lower than 1/3 of the pclk frequency. Sampling with a frequency higher than 1/3 of pclk may lead to incorrect results and functional errors.



For example, in the above figure, although the frequency of x_clk2 is 1/2 of pclk, the synchronized x_clk_syn2 is incorrect due to metastability issues. On the other hand, the frequency of x_clk3 is 1/3 of pclk, and the synchronized x_clk_syn3 will have a frequency ranging between $1/(Tx_clk+Tpclk)$ and $1/(Tx_clk-Tpclk)$. Therefore, if you need to debounce the gpio_in data when generating an interrupt, it is recommended to configure the dfg register frequency splitting PCLK in order to achieve the desired debounce effect, where the period of the debounce clock should be an integer multiple of pclk.

10.2. GATE COUNT

The gate count of GPIO is as the table below.

lib:

GF22FDX_SC7P5T_116CPP_BASE_CSC36L_TT_0P80V_0P00V_0P00V_0P00V_25C

NAND Gate Unit Name: SC7P5T_ND2X1_L_CSC20L

NAND Gate Unit Area: 0.20880

Table 10-1 Gate Count for GPIO QM

GPIO_BUS_WIDTH	gate count
1	1787
$n(n \geq 1)$	$1787 + 234.09 * (n - 1)$
32	9044

11. Function safety

VSI_GPIO_Fusa_Spec_v1.0.doc for details on how to use GPIO's fusa feature.